MIL-C-60235A(MU)
15 April 1974
SUPERSEDING
MIL-C-60235(MU)
15 June 1966

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, M91 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Impulse, M91, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS 28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Armament Command

D8597230	- Cartridge, Impulse, M91, Assembly
C8837884	- Carton, Packing, Ammunition, for Cartridge, Impulse, M91, M163, M173, or XM201
C8837885	- Box, Packing, Ammunition, for Cartridge, Impulse, M91, M163, M173, or XM201
C9227983	- Box, Fiberboard (Outer) Alternative for Cartridge Impulse, M91
IEL 8597230	- Index of Inspection Equipment List for Cartridge, Impulse, M91

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1 N2 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings, referenced specifications and the following:

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of an disintegration of the primer, propellant, or igniter or derangement of any component.

3.4 Functional requirements.

3.4.1 The cartridge shall not fail to fire.

3.4.2 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 16 inch ounces in a sectioned cartridge, void of igniter and propellant charge.

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3.4.3 Functioning.-The cartridge when tested in an M27 or M99 Initiator shall produce the pressures and ignition delay at the temperatures specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size no. 4, MS 28741, and a pressure chamber with a volume of 1.00 cubic inch.

TABLE I

Temperature	Pressure Pounds per square inch (psi) Min.	Max.	Ignition Delay Milliseconds Max (see 6.2)
-65°F to -70°F	500	1425	65
65°F to 75°F	600	1550	65
160°F to 165°F	650	1600	65

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, laminations, perforations, porosity or any similar defect.

3.5.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 cartridges for primer sensitivity, 30 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistic sample), and 10 sets of unassembled components for dimensional and visual examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance of 3.4. The sample for firing shall consist of the following:

TABLE II

Temperature	Not Vibrated	Vibrated
-65°F to -70°F	6	4
65°F to 75°F	6	4
160°F to 165°F	6	4

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Presentation of lot.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled units and shall include:

4.3.1.2.1 Components produced by a homogeneous process.

4.3.1.2.2 Propellant from not more than one lot.

4.3.1.2.3 Igniter from not more than one lot.

4.3.1.2.4 Primers from not more than one lot.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in Table III and IV. For an individual characteristic, the criteria in Table III shall be used. For total characteristics within a major or minor group, the criteria in Table IV shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in ANSI B46.1 N2 shall be used as a basis of comparison for surface roughness determination.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE IV

Total Allowable Defects Per Sample Size

Sample Size for Major Defects - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

Sample Size for Minor Defects - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

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4.3.2.1 Classification of defects.

4.3.2.1.1 Cartridge, Initiator, M91, Assembly, Dwg. D8597230.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray/Visual 1/
2. Improper assembly	X-ray/Visual 1/
3. Mixed ammunition	Visual
4. Metal defective	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge profile failure requiring more than 20 pounds deadweight to insert in concentricity receiver gage	Gage
102. Diameter of head, max.	Gage
103. Diameter of body, max.	Gage
104. Diameter of head recess, min.	Gage
105. Length of datum diameter, max.	Gage
106. Depth of face to head	Gage
107. Dent, failure to chamber in concentricity receiver gage with 20 pounds deadweight	Gage/Visual
108. Firing pin or retaining cup above flush of head face	Visual
109. Foreign matter	Visual
110. Burr	Visual
MINOR	
201. Total length, max.	Gage
202. Dent, which will chamber in concentricity receiver gage with 20 pounds deadweight, max.	Gage/Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Varnish inadequate or missing	Visual
206. Surface finish improper	Visual

1/ The cartridge shall be X-rayed with the long axis of the cartridge perpendicular to the rays of an X-ray machine approved by the contracting officer. The X-ray negatives, properly identified, shall be forwarded to the procuring agency.

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4.3.2.1.2 Carton, Packing, Ammunition, for Cartridge, Initiator, M91, Dwg. C8837884.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual/Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damaged through impregnated box	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.1.3 Box, Packing, Ammunition, for Cartridge, Initiator, M91, Dwg. C8837885.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual/Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of cartons incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

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4.3.2.1.4 Box, Fiberboard (Outer) Alternative, Drawing C9227983.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual
202.	Box broken or damaged	Visual
203.	Number of units contained incorrect	Visual
204.	Marking missing, incorrect or illegible	Visual
205.	Tape incomplete	Visual

4.3.3 Primer examination.-Prior to assembly into a cartridge, each primer shall be visually examined for any discoloration of the foiling paper or a malformation of any kind. Any discoloration of the foiling paper or malformation or failure to comply with the applicable drawing and referenced specification shall be cause for rejection of the primer.

4.3.4 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4:

4.3.4.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of acceptability of coated parts submitted by the contractor shall be in accordance with provisions set forth in Tables III and IV and MIL-C-60539.

4.3.4.2 Primer sensitivity.-A sample of 50 assembled cartridges from the initial production sample and 25 cartridges from each subsequent lot shall be tested for sensitivity. Failure of the sample to comply with the requirement of 3.4.2 shall be cause for rejection of the lot.

4.3.4.3 Airtightness.-Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the requirements of the applicable drawing shall be cause for rejection of the cartridge.

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4.3.4.4 Vibration test, initial production sample.-Twelve of the 30 assembled x-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.3. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in note 1/ of 4.3.2.1.1. Evidence of any disintegration of the primer, propellant or igniter or derangement of any component shall be cause for rejection of the lot. When it becomes necessary to make a change in the lot of primers, propellant, or igniter after the initiation of production, 12 assembled cartridges from the first production lot thereafter shall be tested in accordance with 4.3.4.4.

4.3.4.5 Hardness.-Ten units from each heat-treated lot of components shall be tested. Failure of any unit to comply with the requirements of the drawings shall be cause for rejection of the lot of components.

4.3.4.6 Functional.-The following test shall be performed by the Government:

4.3.4.6.1 Ballistics.-Thirty cartridges for the initial production sample, and 30 cartridges from each production lot, shall be grouped and temperature conditioned as prescribed in Table V and 4.4.6.1.1 and then tested in accordance with 4.4.6. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for lot rejection.

TABLE V

Temperature	Initial Production Sample	Production Lot Sample
-65°F to -70°F	10	10
65°F to 75°F	10	10
160°F to 165°F	10	10

4.3.4.7 Functioning.-The 30 test cartridges, including the 12 vibrated cartridges of the initial production sample (or subsequent sample of 12 test cartridges from the production lot, when a change has been made in the lot of primers, propellant or igniter) and 30 test cartridges from each subsequent lot shall be temperature conditioned in accordance with 4.4.6.1.1 and functionally tested as directed in 4.4.6. Failure of any cartridge to comply with the requirement of 3.4 shall be cause for rejection of the lot.

4.3.5 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8597230.

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4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The anodized part shall be sectioned longitudinally to better expose the critical surfaces to the salt spray test.

4.4.2 Primer sensitivity.-Remove and discard all powder charges from the test cartridges by sectioning the cartridge case around its periphery 1/2 inch, maximu, above the base. Drop test 50 sectioned cartridges from the initial production lot sample and 25 sectioned cartridges from each production lot sample (all void of powder charge) at eight inches, using a steel ball 15/16 inch in diameter, weighing 1.94 ounces plus or minus 0.02 ounce, in a primer sensitivity machine with a cartridge holder approved by the procuring agency.

4.4.3 Airtightness.-The method for determining airtightness shall be by use of the equipment listed on IEL 8597230.

4.4.4 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.5 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.6 Functional tests.-The following tests will be performed by the Government:

4.4.6.1 Ballistics.-The cartridge shall be loaded into acceptable M27 or M99 Initiators with end caps to cover the gas ports during the conditioning period. The cartridges shall be divided into 3 equal groups and temperature conditioned in accordance with Table V. The cartridge shall be tested 5 min after removal from the conditioning chamber. When this time is exceeded, the remover shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.-Cartridges shall be loaded into acceptable M27 or M99 Initiators with end caps to cover gas ports during the conditioning period. The temperature of the propellant within the cartridge shall be considered as the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiators undergoing tests. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring agency. Level A: Drawings C8837884 and C8837885.
Level B: Drawings C8837884 and C9227983.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the title, number and date of this specification.

6.2 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Cartridge Assembly Hazard Data:
Explosive Hazard Classification - Class 3
Storage Compatibility - Group B
Department of Transportation (DOT) Marking - Class C
DOT Marking - Small Arms Ammunition

6.4 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project no. 1377-A311

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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